

Linear Algebra and Several Variable Calculus Notes

Angus

2026.9.22

Abstract

This is a simple abstract for the notes.

lecturer Chris Stock, rm2606, cstock@ed.ac.uk

1 information

20% participation/workshop, 80% final exam.

- solve Linear equations
- define vectors/ dot product
- systems of equations
- matrixs

1.1 work with real numbers

introduce $\mathbb{R}^n$ (actually a vector space)

1.1.1 nature of $\mathbb{R}$

if $x, y \in \mathbb{R}$, then

$$\begin{aligned}xy &= yx \quad (\text{for matrices this may not be true}) \\x + y &= y + x \\-x &\in \mathbb{R} \\x^{-1} &\in \mathbb{R} \quad (x \neq 0)\end{aligned}$$

1.1.2 define points in space

$$\begin{aligned}\dim = 2 &: \mathbb{R}^2, (x, y) \\ \dim = 3 &: \mathbb{R}^3, (x, y, z) \\ \dim = n &: \mathbb{R}^n, (x_1, x_2, \dots, x_n)\end{aligned}$$

1.1.3 addition and subtraction in $\mathbb{R}^n$

in $\mathbb{R}^n$ we can add and subtract vectors
for example

$$A = (a_1, a_2, \dots, a_n), B = (b_1, b_2, \dots, b_n) \\ A + B = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

1.1.4 scalar multiplication

Let c be a scalar in $\mathbb{R}$. The definition is

$$cA = (ca_1, ca_2, \dots, ca_n).$$

1.2 vectors

vectors in $\mathbb{R}^3$ is Ordered pairs "pairs"

vector: from A point to B point, we can easily move them around

$$\overrightarrow{AB} = B - A$$

$\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel if

$$\overrightarrow{AB} = k\overrightarrow{CD}$$

1.2.1 representation of vectors

We have two way to represent vectors.

row vector: $[x_1, x_2, \dots, x_n]$ vectors in row space

column vector: $[x_1, x_2, \dots, x_n]^T$ vectors in column space

to understand how the rows and columns are related to each other, we have fundamentum theorem of linear algebra.

row space and column space have the same dimension

so row rank are equals to column rank.

1.2.2 define dot product

Dot product is a useful tool to describe different relations related to parallelism
Assume

$$A = (a_1, a_2, \dots, a_n), \quad B = (b_1, b_2, \dots, b_n) \\ A \cdot B = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

A and B are vectors in $\mathbb{R}^n$

length: $||A|| = \sqrt{A \cdot A} = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$ is a real number

perpendicular: $A \cdot B = 0$

For understanding dot product, consider A,B. First project A to B at point P, c is a scalar.

$$\overrightarrow{P} = c \overrightarrow{B}$$

And

$$\overrightarrow{AP} = A - cB$$

Then because $\overrightarrow{AP}$ is perpendicular to B .

$$\begin{aligned}(A - cB) \cdot B &= 0 \\ A \cdot B - cB \cdot B &= 0 \\ A \cdot B - c\|B\|^2 &= 0 \\ c &= \frac{A \cdot B}{\|B\|^2}\end{aligned}$$

Dot product is a projection of A onto B resulting in a scalar c .

And we have $\cos(\theta)$

$$\cos(\theta) = \frac{c \|B\|}{\|A\|}$$

So combining the equations gives

$$A \cdot B = \|A\| \|B\| \cos(\theta)$$

The meaning of dot product is calculate how much of A parallel to B And think what you are working with: scalar or vector.

1.3 lines and planes

1.3.1 consider lines

for a vector $X = P + tA$, where P and A are vectors in $\mathbb{R}^n$ and t is a scalar
consider a line in $\mathbb{R}^2$. $X = (x, y) = (p, q) + t(a, b)$ is two equations

$$\begin{aligned}x &= p + ta \\ y &= q + tb\end{aligned}$$

Change the first and the second equation.

$$\begin{aligned}bx &= bp + tab \\ ay &= aq + tab\end{aligned}$$

subtraction two equations.

$$bx - ay = bp - aq$$

times $\frac{1}{a}$

$$\begin{aligned}\frac{b}{a}x - y &= \frac{bp - aq}{a} \\ y &= \frac{b}{a}x - \frac{bp - aq}{a} \\ y &= mx + c\end{aligned}$$

is the non-vector equation of the line.

1.3.2 planes

Consider a plane in $\mathbb{R}^3$, defined by a vector N perpendicular to the plane and a point P in the plane.

Assume $X = (x, y, z), N = (a, b, c)$

$$\begin{aligned}(X - P) \cdot N &= 0 && \text{(perpendicular)} \\ ax + by + cz - N \cdot P &= 0 && N \cdot P \text{ is a scalar} \\ ax + by + cz &= d\end{aligned}$$

1.4 matrices

definition of matrix originally from equations.

$$\begin{aligned}2x - y &= 0 \\ -x + 2y &= 3\end{aligned}$$

And for matrix

$$\begin{aligned}A &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \\ \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} \\ AX &= B\end{aligned}$$